

1. Original Data

BIN

Column 1	date_time	lot_id	wafer_id	item_id	subitem_id	chip_x	chip_y	value
0	11/21/2024 4:...	AAA.1	1	BIN	1	-1.00	3.00	1.00
1	11/21/2024 4:...	AAA.1	1	BIN	2	0.00	3.00	1.00
2	11/21/2024 4:...	AAA.1	1	BIN	3	1.00	3.00	2.00
3	11/21/2024 4:...	AAA.1	1	BIN	4	-3.00	2.00	1.00
4	11/21/2024 4:...	AAA.1	1	BIN	5	-2.00	2.00	2.00
5	11/21/2024 4:...	AAA.1	1	BIN	6	-1.00	2.00	2.00
6	11/21/2024 4:...	AAA.1	1	BIN	7	0.00	2.00	2.00
7	11/21/2024 4:...	AAA.1	1	BIN	8	1.00	2.00	2.00
8	11/21/2024 4:...	AAA.1	1	BIN	9	2.00	2.00	2.00
9	11/21/2024 4:...	AAA.1	1	BIN	10	3.00	2.00	1.00
10	11/21/2024 4:...	AAA.1	1	BIN	11	-3.00	1.00	1.00
11	11/21/2024 4:...	AAA.1	1	BIN	12	-2.00	1.00	1.00
12	11/21/2024 4:...	AAA.1	1	BIN	13	-1.00	1.00	1.00
13	11/21/2024 4:...	AAA.1	1	BIN	14	0.00	1.00	1.00
14	11/21/2024 4:...	AAA.1	1	BIN	15	1.00	1.00	4.00
15	11/21/2024 4:...	AAA.1	1	BIN	16	2.00	1.00	1.00
16	11/21/2024 4:...	AAA.1	1	BIN	17	3.00	1.00	1.00
17	11/21/2024 4:...	AAA.1	1	BIN	18	-4.00	0.00	101.00
18	11/21/2024 4:...	AAA.1	1	BIN	19	-3.00	0.00	102.00
19	11/21/2024 4:...	AAA.1	1	BIN	20	2.00	0.00	1.00

TRACKING

Column 1	date_time	lot_id	wafer_id	step_sequence	step_descripti...	eqp_id	recipe_id
0	2024-10-13 16:...	AAA.1	1	3K108010	10.8 Ch.hole p...	WSO103	SC100
1	2024-10-13 16:...	AAA.1	2	3K108010	10.8 Ch.hole p...	WSO103	SC100
2	2024-10-13 16:...	AAA.1	3	3K108010	10.8 Ch.hole p...	WSO103	SC100
3	2024-10-13 16:...	AAA.1	4	3K108010	10.8 Ch.hole p...	WSO103	SC100
4	2024-10-13 16:...	AAA.1	5	3K108010	10.8 Ch.hole p...	WSO103	SC100
5	2024-10-13 16:...	AAA.1	6	3K108010	10.8 Ch.hole p...	WSO103	SC100
6	2024-10-13 16:...	AAA.1	7	3K108010	10.8 Ch.hole p...	WSO103	SC100
7	2024-10-13 17:...	AAA.1	8	3K108010	10.8 Ch.hole p...	WSO103	SC100
8	2024-10-13 17:...	AAA.1	9	3K108010	10.8 Ch.hole p...	WSO103	SC100
9	2024-10-13 17:...	AAA.1	10	3K108010	10.8 Ch.hole p...	WSO103	SC100
10	2024-10-13 17:...	AAA.1	11	3K108010	10.8 Ch.hole p...	WSO103	SC100
11	2024-10-13 17:...	AAA.1	12	3K108010	10.8 Ch.hole p...	WSO103	SC100
12	2024-10-13 17:...	AAA.1	13	3K108010	10.8 Ch.hole p...	WSO103	SC100
13	2024-10-13 17:...	AAA.1	14	3K108010	10.8 Ch.hole p...	WSO103	SC100
14	2024-10-13 17:...	AAA.1	15	3K108010	10.8 Ch.hole p...	WSO103	SC100
15	2024-10-13 17:...	AAA.1	16	3K108010	10.8 Ch.hole p...	WSO103	SC100
16	2024-10-13 17:...	AAA.1	17	3K108010	10.8 Ch.hole p...	WSO103	SC100
17	2024-10-13 17:...	AAA.1	18	3K108010	10.8 Ch.hole p...	WSO103	SC100
18	2024-10-13 17:...	AAA.1	19	3K108010	10.8 Ch.hole p...	WSO103	SC100

2. Integrated Data

BIN_T/F

lot_id	wafer_id	item_id	subitem_id	chip_x	chip_y	value	Determination	T/F
AAA.1	1	BIN	1	-1.00	3.00	1.00	TRUE	1
AAA.1	1	BIN	2	0.00	3.00	1.00	TRUE	1
AAA.1	1	BIN	3	1.00	3.00	2.00	TRUE	1
AAA.1	1	BIN	4	-3.00	2.00	1.00	TRUE	1
AAA.1	1	BIN	5	-2.00	2.00	2.00	TRUE	1
AAA.1	1	BIN	6	-1.00	2.00	2.00	TRUE	1
AAA.1	1	BIN	7	0.00	2.00	2.00	TRUE	1
AAA.1	1	BIN	8	1.00	2.00	2.00	TRUE	1
AAA.1	1	BIN	9	2.00	2.00	2.00	TRUE	1
AAA.1	1	BIN	10	3.00	2.00	1.00	TRUE	1
AAA.1	1	BIN	11	-3.00	1.00	1.00	TRUE	1
AAA.1	1	BIN	12	-2.00	1.00	1.00	TRUE	1
AAA.1	1	BIN	13	-1.00	1.00	1.00	TRUE	1
AAA.1	1	BIN	14	0.00	1.00	1.00	TRUE	1
AAA.1	1	BIN	15	1.00	1.00	4.00	FALSE	0
AAA.1	1	BIN	16	2.00	1.00	1.00	TRUE	1
AAA.1	1	BIN	17	3.00	1.00	1.00	TRUE	1
AAA.1	1	BIN	18	-4.00	0.00	101.00	FALSE	0
AAA.1	1	BIN	19	-3.00	0.00	102.00	FALSE	0
AAA.1	1	BIN	20	-2.00	0.00	1.00	TRUE	1
AAA.1	1	BIN	21	-1.00	0.00	1.00	TRUE	1
AAA.1	1	BIN	22	0.00	0.00	1.00	TRUE	1
AAA.1	1	BIN	23	1.00	0.00	1.00	TRUE	1
AAA.1	1	BIN	24	2.00	0.00	2.00	TRUE	1
AAA.1	1	BIN	25	3.00	0.00	1.00	TRUE	1
AAA.1	1	BIN	26	4.00	0.00	1.00	TRUE	1
AAA.1	1	BIN	27	-3.00	-1.00	1.00	TRUE	1
AAA.1	1	BIN	28	-2.00	-1.00	1.00	TRUE	1
AAA.1	1	BIN	29	-1.00	-1.00	104.00	FALSE	0
AAA.1	1	BIN	30	0.00	-1.00	1.00	TRUE	1
AAA.1	1	BIN	31	1.00	-1.00	1.00	TRUE	1
AAA.1	1	BIN	32	2.00	-1.00	101.00	FALSE	0
AAA.1	1	BIN	33	3.00	-1.00	1.00	TRUE	1
AAA.1	1	BIN	34	-3.00	-2.00	1.00	TRUE	1
AAA.1	1	BIN	35	-2.00	-2.00	101.00	FALSE	0
AAA.1	1	BIN	36	-1.00	-2.00	1.00	TRUE	1
AAA.1	1	BIN	37	0.00	-2.00	1.00	TRUE	1
AAA.1	1	BIN	38	1.00	-2.00	1.00	TRUE	1
AAA.1	1	BIN	39	2.00	-2.00	1.00	TRUE	1
AAA.1	1	BIN	40	3.00	-2.00	1.00	TRUE	1
AAA.1	1	BIN	41	-1.00	-3.00	1.00	TRUE	1
AAA.1	1	BIN	42	0.00	-3.00	1.00	TRUE	1
AAA.1	1	BIN	43	1.00	-3.00	1.00	TRUE	1

BIN_Yield

lot_id	wafer_id	양호 count	Yield
AAA.1	1	37.00	86.05
AAA.1	2	37.00	86.05
AAA.1	3	37.00	86.05
AAA.1	4	37.00	86.05
AAA.1	5	37.00	86.05
AAA.1	6	37.00	86.05
AAA.1	7	36.00	83.72
AAA.1	8	37.00	86.05
AAA.1	9	37.00	86.05
AAA.1	10	36.00	83.72
AAA.1	11	35.00	81.40
AAA.1	12	37.00	86.05
AAA.1	13	36.00	83.72
AAA.1	14	37.00	86.05
AAA.1	15	36.00	83.72
AAA.1	16	36.00	83.72
AAA.1	17	36.00	83.72
AAA.1	18	36.00	83.72
AAA.1	19	37.00	86.05
AAA.1	20	36.00	83.72
AAA.1	21	36.00	83.72
AAA.1	22	37.00	86.05
AAA.1	23	37.00	86.05
AAA.1	24	37.00	86.05
AAA.1	25	36.00	83.72
AAB.1	1	37.00	86.05
AAB.1	2	37.00	86.05
AAB.1	3	37.00	86.05
AAB.1	4	36.00	83.72
AAB.1	5	37.00	86.05
AAB.1	6	37.00	86.05
AAB.1	7	36.00	83.72
AAB.1	8	35.00	81.40
AAB.1	9	36.00	83.72
AAB.1	10	37.00	86.05
AAB.1	11	37.00	86.05
AAB.1	12	37.00	86.05
AAB.1	13	36.00	83.72
AAB.1	14	37.00	86.05
AAB.1	15	36.00	83.72
AAB.1	16	36.00	83.72
AAB.1	17	36.00	83.72
AAB.1	18	37.00	86.05

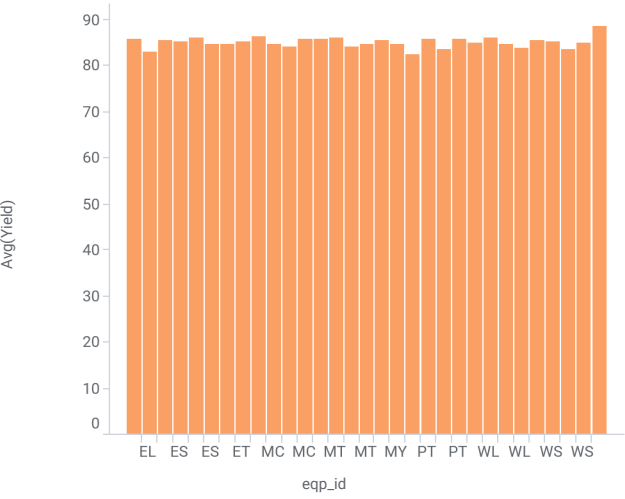
3. Handled Data

Handled + Integrated Data

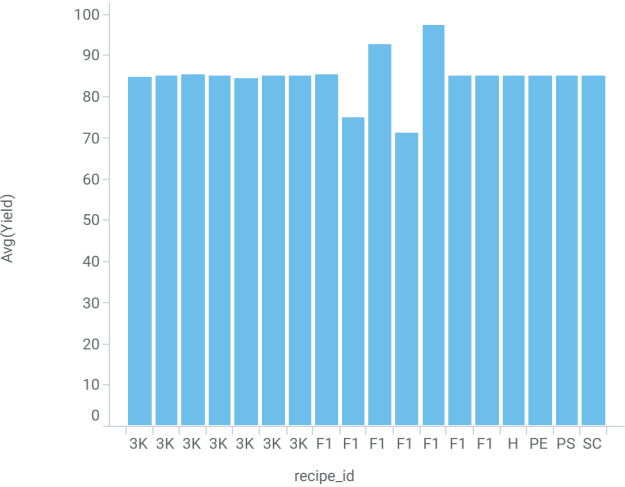
Column 1	date_time	lot_id	wafer_id	step_sequence	step_descripti...	eqp_id	recipe_id	양호 count	Yield
0	2024-10-13 16...	AAA.1	1	3K108010	10.8 Ch.hole p...	WSO103	SC100	37.00	86.05
1	2024-10-13 16...	AAA.1	2	3K108010	10.8 Ch.hole p...	WSO103	SC100	37.00	86.05
2	2024-10-13 16...	AAA.1	3	3K108010	10.8 Ch.hole p...	WSO103	SC100	37.00	86.05
3	2024-10-13 16...	AAA.1	4	3K108010	10.8 Ch.hole p...	WSO103	SC100	37.00	86.05
4	2024-10-13 16...	AAA.1	5	3K108010	10.8 Ch.hole p...	WSO103	SC100	37.00	86.05
5	2024-10-13 16...	AAA.1	6	3K108010	10.8 Ch.hole p...	WSO103	SC100	37.00	86.05
6	2024-10-13 16...	AAA.1	7	3K108010	10.8 Ch.hole p...	WSO103	SC100	36.00	83.72
7	2024-10-13 17...	AAA.1	8	3K108010	10.8 Ch.hole p...	WSO103	SC100	37.00	86.05
8	2024-10-13 17...	AAA.1	9	3K108010	10.8 Ch.hole p...	WSO103	SC100	37.00	86.05
9	2024-10-13 17...	AAA.1	10	3K108010	10.8 Ch.hole p...	WSO103	SC100	36.00	83.72
10	2024-10-13 17...	AAA.1	11	3K108010	10.8 Ch.hole p...	WSO103	SC100	35.00	81.40
11	2024-10-13 17...	AAA.1	12	3K108010	10.8 Ch.hole p...	WSO103	SC100	37.00	86.05
12	2024-10-13 17...	AAA.1	13	3K108010	10.8 Ch.hole p...	WSO103	SC100	36.00	83.72
13	2024-10-13 17...	AAA.1	14	3K108010	10.8 Ch.hole p...	WSO103	SC100	37.00	86.05
14	2024-10-13 17...	AAA.1	15	3K108010	10.8 Ch.hole p...	WSO103	SC100	36.00	83.72
15	2024-10-13 17...	AAA.1	16	3K108010	10.8 Ch.hole p...	WSO103	SC100	36.00	83.72
16	2024-10-13 17...	AAA.1	17	3K108010	10.8 Ch.hole p...	WSO103	SC100	36.00	83.72
17	2024-10-13 17...	AAA.1	18	3K108010	10.8 Ch.hole p...	WSO103	SC100	36.00	83.72
18	2024-10-13 17...	AAA.1	19	3K108010	10.8 Ch.hole p...	WSO103	SC100	37.00	86.05
19	2024-10-13 17...	AAA.1	20	3K108010	10.8 Ch.hole p...	WSO103	SC100	36.00	83.72
20	2024-10-13 17...	AAA.1	21	3K108010	10.8 Ch.hole p...	WSO103	SC100	36.00	83.72
21	2024-10-13 17...	AAA.1	22	3K108010	10.8 Ch.hole p...	WSO103	SC100	37.00	86.05
22	2024-10-13 17...	AAA.1	23	3K108010	10.8 Ch.hole p...	WSO103	SC100	37.00	86.05
23	2024-10-13 17...	AAA.1	24	3K108010	10.8 Ch.hole p...	WSO103	SC100	37.00	86.05
24	2024-10-13 17...	AAA.1	25	3K108010	10.8 Ch.hole p...	WSO103	SC100	36.00	83.72
25	2024-10-13 18...	AAA.1	1	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	37.00	86.05
26	2024-10-13 18...	AAA.1	2	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	37.00	86.05
27	2024-10-13 18...	AAA.1	3	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	37.00	86.05
28	2024-10-13 18...	AAA.1	4	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	37.00	86.05
29	2024-10-13 18...	AAA.1	5	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	37.00	86.05
30	2024-10-13 18...	AAA.1	6	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	37.00	86.05
31	2024-10-13 18...	AAA.1	7	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	36.00	83.72
32	2024-10-13 18...	AAA.1	8	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	37.00	86.05
33	2024-10-13 18...	AAA.1	9	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	37.00	86.05
34	2024-10-13 19...	AAA.1	10	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	36.00	83.72
35	2024-10-13 19...	AAA.1	11	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	35.00	81.40
36	2024-10-13 19...	AAA.1	12	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	37.00	86.05
37	2024-10-13 19...	AAA.1	13	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	36.00	83.72
38	2024-10-13 19...	AAA.1	14	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	37.00	86.05
39	2024-10-13 19...	AAA.1	15	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	36.00	83.72
40	2024-10-13 19...	AAA.1	16	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	36.00	83.72
41	2024-10-13 19...	AAA.1	17	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	36.00	83.72
42	2024-10-13 19...	AAA.1	18	3K108100	10.8 Ch.hole ...	TLO101	PESION2000	36.00	83.72

4. Yield Data Visualization

Yield per eqp_id



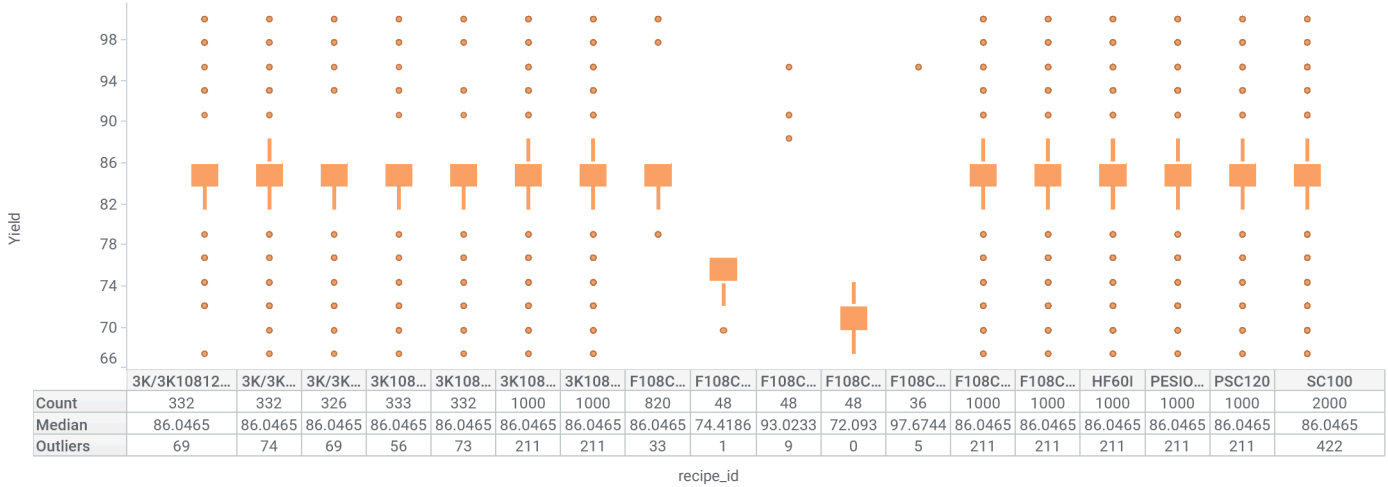
Yield per recipe_id



Data Relationships (Anova)

Y (numerical)	X (categorical)	p-value ▲	FStat ▼	S2Btwn	S2Wthn	dfBtwn	dfWthn	n
Yield	recipe_id	6.77E-297	742.65	22620.33	7576.66	4	995	1000
Yield	eqp_id	1.26E-009	20.92	1216.31	28980.68	2	997	1000

Data Relationships (Details)



5. Path and Yield

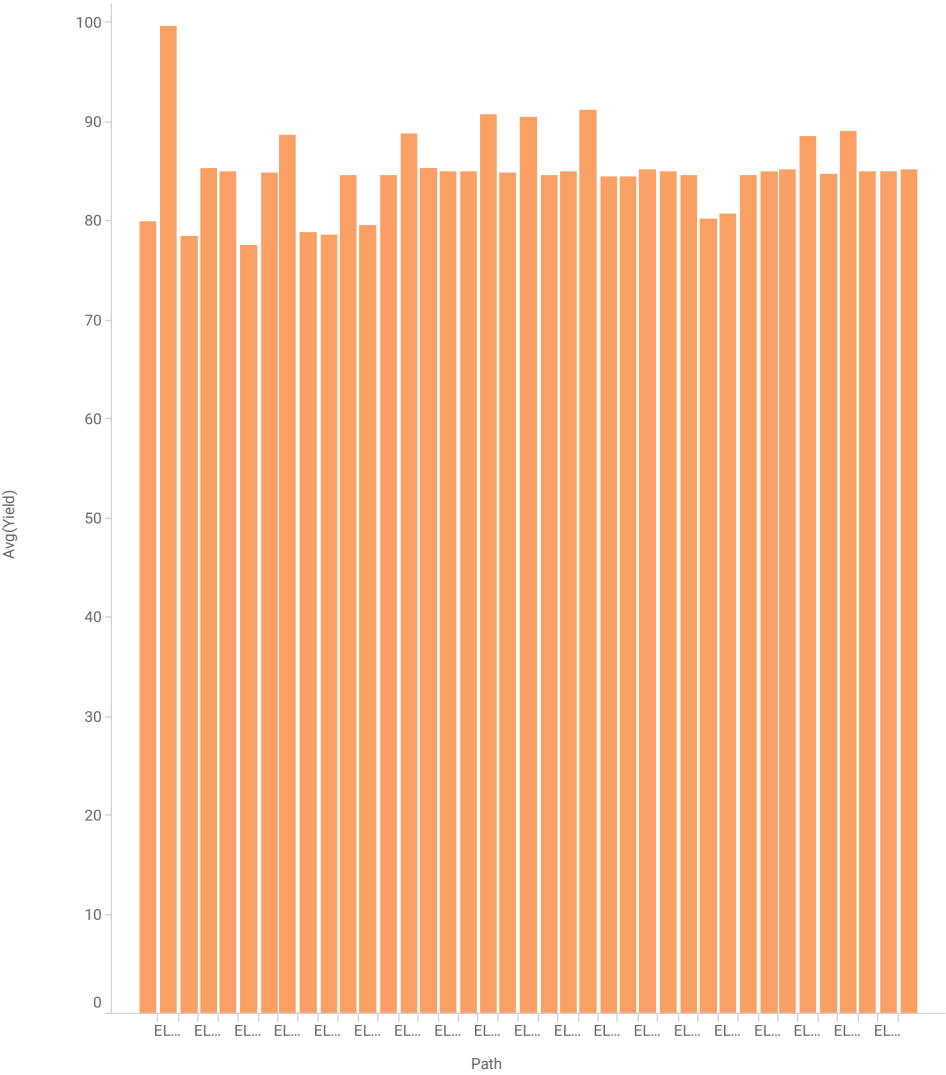
PATH_INTEGRATED

[illegible]

Data table:
PATH_INTEGRATED

6. Path and Yield Visualization

Yield per Path



Data Relationships (2) (Anova)

Y (numerical)	X (categorical)	p-value ▲	FStat ▼	S2Btwn	S2Wthn	dfBtwn	dfWthn
Yield	Path	4.88E-133	29.17	16173.74	14023.26	38	961

Data Relationships (2) (Details)

